

IPDS Sensor Node: ESP32 + MPU6050 + DS3231 + NEO-6M

I2C bus: SDA = GPIO21, SCL = GPIO22, 4.7k pull-ups R1/R2 to 3V3

MPU6050 AD0 tied to 3V3 -> I2C address 0x69 (DS3231 RTC uses 0x68)

GPS on UART2 @ 9600 baud: GPIO16 (RX2) <- GPS TX, GPIO17 (TX2) -> GPS RX

Power: 5V into J1 (via D1) or the DevKit USB. Don't use both unless your DevKit has a USB isolation diode.

I2C: GPIO21=SDA GPIO22=SCL | UART2: GPIO16=RX2 GPIO17=TX2
 ESP32-DevKitC (38-pin) + MPU6050 (GY-521) + DS3231 (ZS-042) + NEO-6M (GY-NEO6M)

Intelligent Pothole Detection System

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 File: IPDS_SensorNode.kicad_sch

Title: IPDS Sensor Node

Size: A4	Date: 2026-10-08	Rev: 1.1
KiCad E.D.A. 10.0.6		Id: 1/1

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